

ECG Anomaly Detection — Session Report

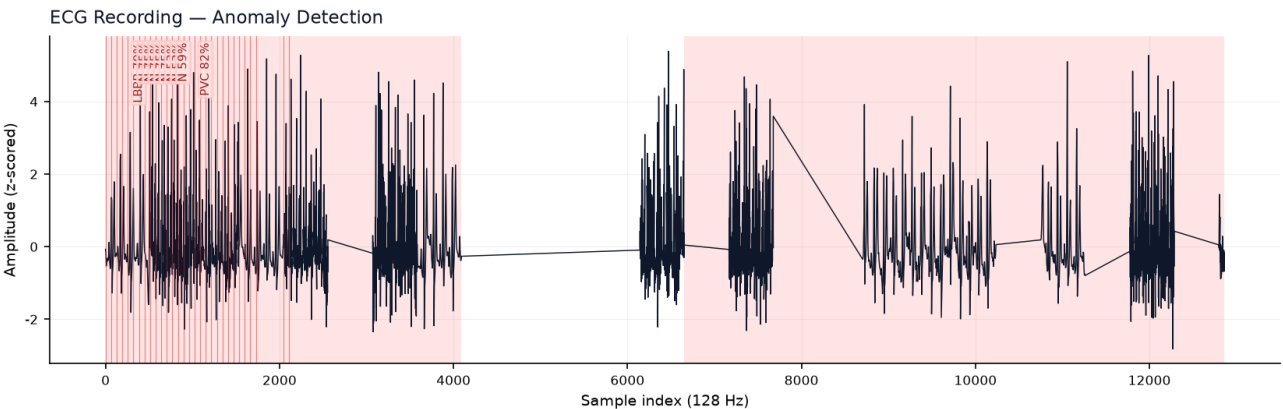
Prepared for: Attending Physician • Session ID: f0e149b7 • Generated: 2026-07-05 07:31:18

Session Information

Patient / User	Sriyantha Sadaruwan
Source	mit-bih-arrhythmia (111)
Started	2026-07-05 07:31:18
Ended	—
Duration	—
Status	running
Total beats analyzed	0
Anomalous beats	0 (0.00%)
Unique alerted beats in report	1,188
Model	ecg-ae-v2 (v1783189189) • Dilated-U-Net-Autoencoder
Analysis mode	Sliding autoencoder reconstruction
Detection threshold (MAE)	0.1341

ECG Recording & Detection Results

The following plot shows the recorded ECG signal (z-score normalized). Red-shaded regions indicate samples flagged by autoencoder reconstruction error.



Main Arrhythmia Finding

Left Bundle Branch Block was the main high-confidence arrhythmia type detected in this session. It appeared 252 time(s) with average confidence 72.7% and maximum confidence 94.7%.

Reason: the autoencoder first flagged the ECG segment by reconstruction error, and the transferred classifier then assigned this label with at least 50% confidence. Lower-confidence labels are excluded to avoid over-interpreting uncertain model output.

Example: time 2026-07-05 07:35:08, anomaly score 0.0636, threshold 0.0500, classifier confidence 94.7%.

Clinical meaning: Left Bundle Branch Block (LBBB) is a condition where electrical signals to the left ventricle of the heart are delayed. This causes the left ventricle to contract later than the right ventricle, resulting in an abnormal ECG pattern with a wide QRS complex. LBBB affects the timing of the heartbeat but not necessarily the heart's ability to pump blood.

Why it matters: LBBB itself is not life-threatening, but it often indicates underlying heart disease. New-onset LBBB can sometimes signal an acute heart attack and requires immediate medical evaluation. Chronic LBBB (present for a long time) is usually monitored but may require treatment of the underlying condition.

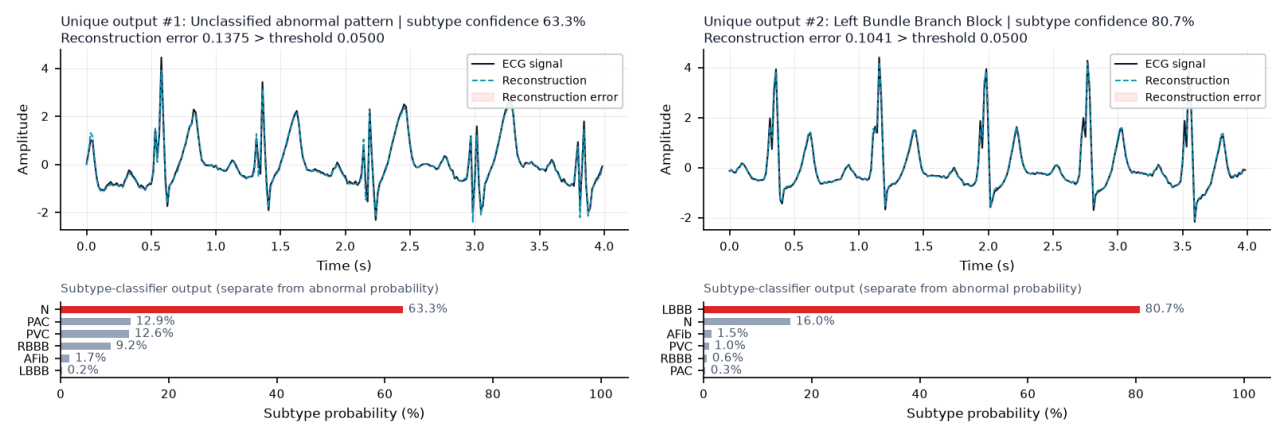
Model Output Summary

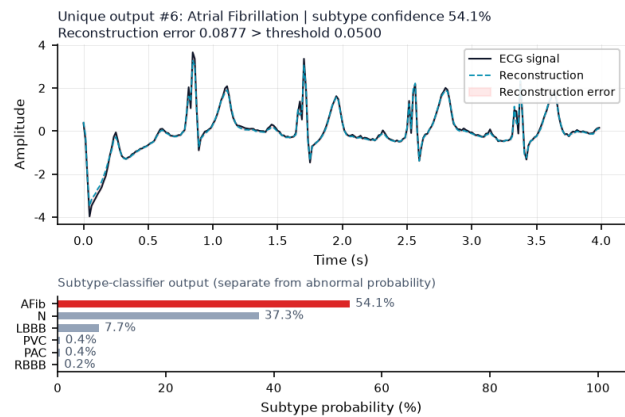
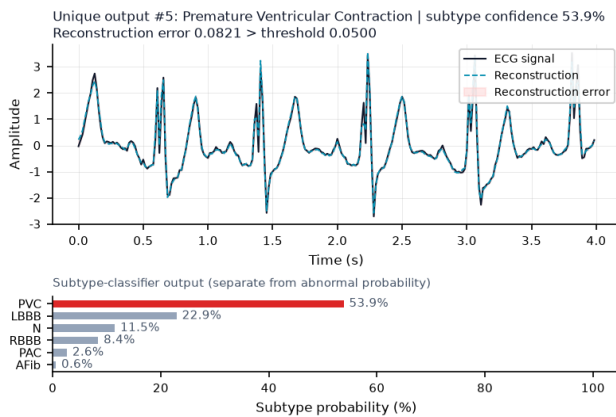
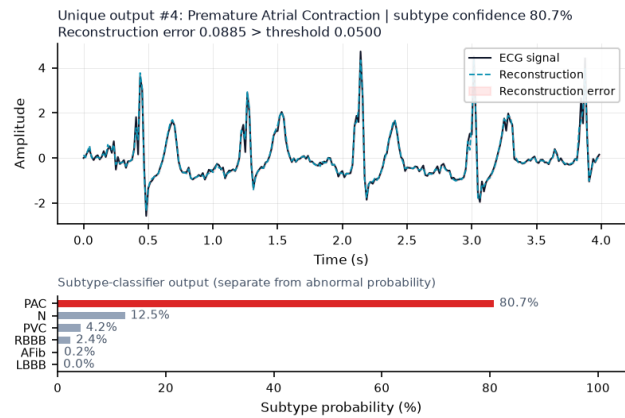
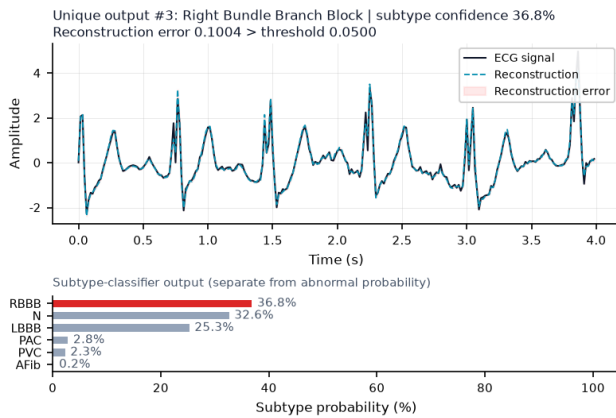
Each row is one distinct model output. Beat counts are calculated after repeated R-peak alerts are removed. The binary decision answers "is this beat abnormal?" The subtype confidence separately describes the predicted abnormal type.

Model output	Beats	Binary decision	Subtype conf.	RR	Input
Unclassified abnormal pattern	628	error 0.1375 > thr. 0.0500	63.3%	—	not recorded
Left Bundle Branch Block	325	error 0.1041 > thr. 0.0500	80.7%	—	not recorded
Right Bundle Branch Block	112	error 0.1004 > thr. 0.0500	36.8%	—	not recorded
Premature Atrial Contraction	56	error 0.0885 > thr. 0.0500	80.7%	—	not recorded
Premature Ventricular Contraction	47	error 0.0821 > thr. 0.0500	53.9%	—	not recorded
Atrial Fibrillation	20	error 0.0877 > thr. 0.0500	54.1%	—	not recorded

Unique Anomaly Waveforms and Model Outputs

Each panel shows one representative segment for a distinct model output, together with reconstruction error, threshold, label, and classifier probabilities. Repeated outputs are not plotted.





Clinical Summary

Fewer than 1% of analysed beats exceeded the model threshold. This research result does not exclude arrhythmia. High-confidence classifier findings in this report include: Left Bundle Branch Block, Right Bundle Branch Block, Premature Atrial Contraction, Premature Ventricular Contraction. Only arrhythmia labels with confidence above 50% are included.

Physician Notes

Physician signature: _____

Date: _____

This report is generated automatically by a research ECG anomaly-detection system and is intended for research review only. Model probability and subtype confidence are not clinical certainty. This report does not constitute a medical diagnosis and must not replace qualified clinical judgment.